

F24-039-D-OptiFit

Project Team

Zain Ul Abideen	21I-0747
Fariz Ahmad	21I-0552
Muhammad Moaz Farrukh	21I-0899

Session 2021-2025

Supervised by

Dr. Imran Ashraf



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

June, 2025

Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Face 3D Model Generation	2
1.3.2	Glasses 3D Model Generation	2
1.3.3	3D Model Integration	3
1.3.4	Glasses Customization	3
1.3.5	API Development	3
1.3.6	Client Web Application Development	3
2	Project Requirements	5
2.1	Use-case/Event Response Table/Storyboarding	5
2.1.1	UseCase Diagram	5
2.1.2	High Level Use-Cases	7
2.1.3	Extended Use-Cases	9
2.2	Functional Requirements	26
2.2.1	Face 3D Model Generation	26
2.2.2	Glasses 3D Model Generation	26
2.2.3	3D Model Integration	26
2.2.4	Glasses Customization	27
2.2.5	API Development	27
2.2.6	Client Web Application Development	27
2.3	Non-Functional Requirements	27
2.3.1	Performance	27
2.3.2	Reliability	28
2.3.3	Usability	28
2.3.4	Security	28
3	System Overview	29
3.1	Architectural Design	30
3.1.1	Architecture Diagram	30
3.2	Design Models	31

3.2.1	Activity Diagram	31
3.2.2	Data Flow Diagram	39
3.2.3	System-level Sequence Diagram	40
3.2.4	ERD Diagram	45
3.3	Data Design	46
References		47

List of Figures

2.1	Use-Case Diagram	6
3.1	Architecture diagram	30
3.2	Detect Face Activity Diagram	31
3.3	Measure Face Dimension Activity Diagram	32
3.4	Generate Glasses Model Activity Diagram	33
3.5	Preview Glasses Frames in 3D Activity Diagram	34
3.6	Fit 3D Frame on Face Activity Diagram	35
3.7	Customize Glasses Frame Activity Diagram	36
3.8	Integrate With Retailer Activity Diagram	37
3.9	Select Glasses Frame Activity Diagram	38
3.10	Data Flow Diagram	39
3.11	Detect Face	40
3.12	Measure Face Dimensions	41
3.13	Generate Glasses Model	42
3.14	Preview Frames in 3D	42
3.15	Fit 3D Frame	43
3.16	Customize Glasses Frame	43
3.17	Integrate with Retailers	44
3.18	Select Glasses Frames	44
3.19	ERD Diagram - OptiFit	45

List of Tables

2.1	Detect Face High Level Use Case	7
2.2	Measure Face Dimensions High Level Use Case	7
2.3	Generate Glasses Model High Level Use Case	7
2.4	Preview Glasses Frames in 3D High Level Use Case	7
2.5	Fit 3D Frame on Face High Level Use Case	8
2.6	Customize Glasses Frame High Level Use Case	8
2.7	Integrate with Retailers High Level Use Case	8
2.8	Select Glasses Frame High Level Use Case	8
2.9	Detect Face Extended Use-Case	10
2.10	Measure Face Dimensions Extended Use-Case	12
2.11	Generate Glasses 3D Model Extended Use-Case	14
2.12	Preview Glasses Frames in 3D Extended Use-Case	16
2.13	Fit 3D Frame Extended Use-Case	18
2.14	Customize Frame Extended Use-Case	20
2.15	Integrate with Retailers Extended Use-Case	23
2.16	Select Frame Glasses Extended Use-Case	25

Chapter 1

Introduction

The Virtual Glasses Try-on solution allows users to try various glasses frames on their face using their device camera in a real-time. Our solution is a service-based model. Our solution imposes the 3d model of the glasses onto the user's face using facial recognition and computer vision technologies, therefore, providing an interactive and customized experience to the users. Our product can easily and quickly integrated into the platforms and systems currently used by other eyewear retailers.

1.1 Problem Statement

When buying glasses physically from a shop, users can have the option to try glasses physically on their face and choose the one that suits their face shape, style, and preferences. But when buying glasses online, it can be a frustrating and uncertain experience for the users. Without physically trying the glasses, users can come up with the following questions:

- Will these glasses feel good on me?
- Will the style align with my personal preferences?
- Will these glasses fit the shape of my face?

Since one cannot try on glasses before purchasing them online, this would lead to an increased number of returns, exchanges, and dissatisfied customers. This results in wasted time and money for both customers and sellers, which highlights the need for a better online glasses shopping experience. This will not only help users to make better choices but also reduce return rates, benefiting both customers and sellers. [3]

1.2 Scope

This project aims to develop an advanced virtual glasses try-on system that uses 3D face and glasses models to let users try on and customize glasses. The Project is divided into various phases, starting from face 3D model generation which focuses on detecting and analyzing facial features to create the 3D mesh of the face. Next, in the Glasses 3D Model Generation module images and videos of glasses are used to produce a 3D mesh of glasses. This involves image segmentation to isolate the glasses, then using photogrammetry to generate a mesh from multiple viewpoints. After that the face and glasses models are merged to produce the mesh of face wearing glasses. Additionally it provides users the ability to personalize the appearance of their glasses by adjusting color and size. To share this capability to the retailers, it will include an API to link it with retailers system. To demo the results, there will be a demo website which will have the API integrated. The website includes a content management system, a 3D frame previewer and the virtual glasses try on feature.

1.3 Modules

1.3.1 Face 3D Model Generation

This module allows us to extract features from the face to generate a 3D mesh.

1. **Face Detection:** Scan the video to locate and identify the face by utilizing different facial features such as eyes, nose, mouth, and jawline.
2. **Feature Extraction:** Identify and categorize facial features using landmark detection, and analyze texture and shape details.
3. **Face Mesh Generation:** Create a 3D model based on the extracted features using predefined models to fit the detected features into a 3D shape.
4. **Face size estimation:** Estimate the size of the users face using depth estimation via diameter of the iris.

1.3.2 Glasses 3D Model Generation

This module involves generating a 3D model of the glasses using images and videos.

1. **Image Segmentation:** Preprocess images to separate the glasses from the background, ensuring that only the glasses are visible.

2. **Glasses Mesh Generation:** Use images from different viewpoints and apply photogrammetry to create a point cloud and a 3D mesh of the glasses.

1.3.3 3D Model Integration

This module integrates the 3D models of the face and glasses to create a cohesive mesh.

1. **Feature Extraction:** Identify common features in both the glasses and face to ensure consistency across different types.
2. **Feature Mapping:** Map the features of the glasses to the features of the face to generate a unified 3D mesh.
3. **Model Integration:** Connect and assimilate the glasses mesh with the face mesh to create a joint model of the face wearing glasses.

1.3.4 Glasses Customization

This module allows users to customize the appearance of the glasses.

1. **Color:** Choose from a variety of colors to personalize the frame of the glasses.
2. **Size:** Adjust the size of the glasses to meet individual preferences.

1.3.5 API Development

Create an Application Programming Interface (API) to connect the main program with retailers' websites and applications.

1. **API Endpoints:** Implement various API endpoints for all functionalities of the virtual glasses try-on system.
2. **API Authentication:** Apply authentication measures to secure all endpoints and ensure authorized access.

1.3.6 Client Web Application Development

Develop a web application to ensure the API functions correctly and to display results to users.

1. **CMS (Content Management System):** Create a system to manage and display glasses to end users.
2. **3D Frame Previewer:** Develop an interactive 3D previewer to show the glasses from all angles.
3. **Glasses Try-On:** Implement functionality for users to try on all available glasses using the integrated API.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of our project.

2.1 Use-case/Event Response Table/Storyboarding

2.1.1 UseCase Diagram

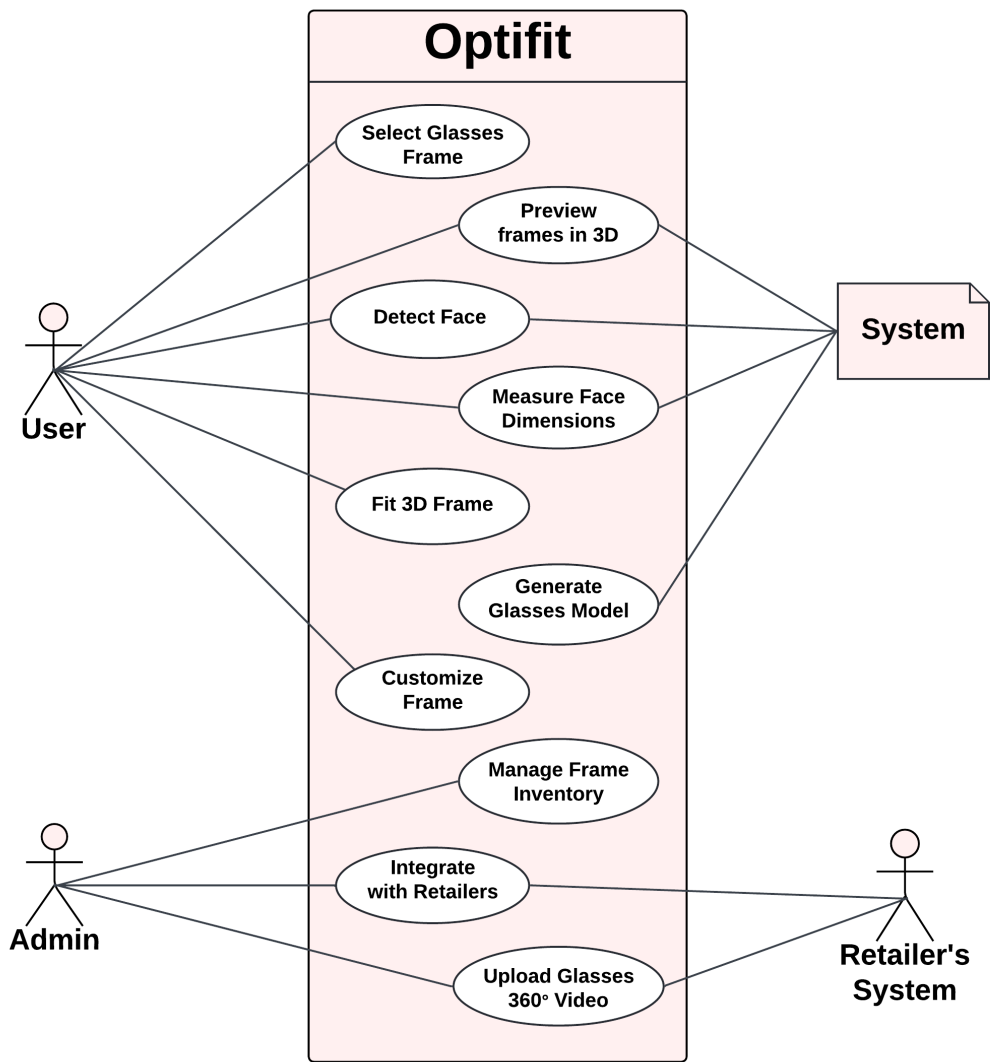


Figure 2.1: Use-Case Diagram

2.1.2 High Level Use-Cases

Use Case	Detect Face
Actors	Customer, System
Type	Primary
Description	The system will detect the customer's face through his / her device's camera in real-time. The customer will allow the camera access, and then the system processes the captured data to recognize facial features in real-time.

Table 2.1: Detect Face High Level Use Case

Use Case	Measure Face Dimensions
Actors	Customer, System
Type	Primary
Description	After detecting the face, the system calculates the facial measurements to ensure proper fitting. Facial measurement includes face width and distance from the camera (depth estimate).

Table 2.2: Measure Face Dimensions High Level Use Case

Use Case	Generate Glasses Model
Actors	Admin, System
Type	Primary
Description	The system will generate a 3D frame model using photogrammetry. In this technique, the admin or user will provide 360-degree video of the glasses. This process also involves image segmentation.

Table 2.3: Generate Glasses Model High Level Use Case

Use Case	Preview Glasses Frames in 3D
Actors	Customer, System
Type	Primary
Description	The system will allow customers to view different glass frames in a 3D environment. Users can rotate and examine the glasses from different angles to get a better sense of the frames.

Table 2.4: Preview Glasses Frames in 3D High Level Use Case

Use Case	Fit 3D Frame on Face
Actors	Customer, System
Type	Primary
Description	The system will overlay the 3D glasses model on the user's face in real-time. It will fit the frame according to the detected face dimensions. This will allow the user to experience the virtual try-on.

Table 2.5: Fit 3D Frame on Face High Level Use Case

Use Case	Customize Glasses Frame
Actors	Customer, System
Type	Primary
Description	The system will allow users to personalize the glasses frame by adjusting its color, size. These customizations will allow users to finalize their selection before purchase.

Table 2.6: Customize Glasses Frame High Level Use Case

Use Case	Integrate with Retailers
Actors	Admin, Retailer's system, System
Type	Secondary
Description	The system will provide APIs that allow retailers to integrate the system with their e-commerce platforms. Admin will manage the integration, ensuring that the system synchronizes correctly with the retailer's system.

Table 2.7: Integrate with Retailers High Level Use Case

Use Case	Select Glasses Frame
Actors	Customer
Type	Primary
Description	The user will browse the collection of available glasses frames and select one. The system will display the frames in different styles, and the user can choose a frame that matches their preferences.

Table 2.8: Select Glasses Frame High Level Use Case

2.1.3 Extended Use-Cases

Use Case	Detect Face
Scope	Facial Recognition System
Level	Functional
Primary Actor	User, System Administrator
Stakeholders and Interests	• User: Accurate face detection of the user. • System Administrator: Efficient and reliable face for system performance. • Developer: Integration with different image sources and facial recognition algorithm.
Pre-conditions	• The user has access to system. • Image capture device (camera) is configured. • A facial recognition system is configured.
Success Guarantee	The system accurately detects users' faces in real-time videos.
Main Success Scenario	• The user initiates the face detection process. • The system captures or receives a video stream. • The facial recognition software processes the stream. • The system detects and highlights faces within the image/stream. • The system stores or displays detection results.

Extensions	• Filter out non-human faces (e.g. animals, cartoons). • Detect multiple faces. • Handle varying lighting conditions.
Special Requirements	• Real-time processing for video streams. • Accuracy > 95%.
Technology and Data Variation	• Video capture devices: webcam, smartphone camera. • Facial recognition algorithms: OpenCV, Deep Learning-based, Mediapipe by Google.

Table 2.9: Detect Face Extended Use-Case

Use Case	Measure Face Dimensions
Scope	Optical measurement system within the virtual try-on platform.
Level	User goal
Primary Actor	User (Customer trying on glasses).
Stakeholders and Interests	• User: Interested in getting accurate face measurements for a perfect fit of glasses. • Eyewear Retailer: Wants to provide a reliable and personalized experience for customers. • Developer: Aims to ensure the face measurement feature is accurate and enhances user satisfaction.

Pre-conditions	• User has provided camera access to the platform. • Face detection and tracking system is initialized and functioning.
Success Guarantee	The system successfully measures key face dimensions (face width, depth estimate) and provides accurate data for glasses fitting.
Main Success Scenario	• User opens the virtual try-on tool. • System requests camera access and user grants permission. • User aligns face in front of the camera following system prompts. • System detects the face and key facial landmarks. • The system calculates face dimensions (face width) using depth estimation and facial points. • The system displays the calculated dimensions to the user or saves it for further steps (e.g. glasses fitting).
Extensions	• Inaccurate face positioning: The system prompts the user to reposition their face until alignment is proper. • The system provides real-time feedback for better positioning.
Special Requirements	• High accuracy for depth and width estimation. • Real-time face tracking to avoid delays in measurement.

Technology and Data Variation	• Uses MediaPipe for face detection and depth estimation algorithms. • Camera data is used to capture facial landmarks.
--------------------------------------	--

Table 2.10: Measure Face Dimensions Extended Use-Case

Use Case	Generate Glasses 3D Model
Scope	3D glasses model creation within the virtual try-on system.
Level	User goal
Primary Actor	User (Retailer or Developer creating custom 3D glasses models).
Stakeholders and Interests	• User (Retailer/Developer): Wants to generate accurate 3D models of glasses for use in the virtual try-on feature. • End Customers: Interested in trying on realistic representations of glasses to visualize fit and style. • Project Owner/Developer: Aims to streamline the model generation process to ensure efficiency and accuracy.
Pre-conditions	• The system has access to glasses images/videos. • The 3D model generation module is set up and functional.
Success Guarantee	The system successfully generates a high-quality accurate 3D model of the glasses ready for virtual try-on.

Main Success Scenario	• User selects the "Generate 3D Glasses Model" option within the platform. • The system prompts the user to upload images/videos for the glasses design. • The system processes the provided input using photogrammetry or segmentation techniques to build a 3D mesh. • The system applies textures and finishes to the generated 3D model. • The user previews the 3D model. • The system saves the finalized 3D model in a usable format (e.g. .obj or .glb). • The model is now ready for integration into the virtual try-on feature.
Extensions	• Input images/videos are unclear or incomplete: The system notifies the user and provides tips on taking better images (e.g. clearer angles, improved lighting). • User uploads new images or videos for processing.
Special Requirements	• High-fidelity 3D model rendering with attention to detail (frame thickness, etc.). • Fast processing times to ensure efficient model generation. • Support for various input formats (images, videos).

Technology and Data Variation	• Uses segmentation and photogrammetry techniques to generate the 3D model from images or videos. • Trimesh or similar 3D libraries for model visualization and customization.
--------------------------------------	---

Table 2.11: Generate Glasses 3D Model Extended Use-Case

Use Case	Preview Frames in 3D
Scope	3D preview functionality within the virtual try-on platform.
Level	User goal
Primary Actor	User (Customer trying to preview glasses in 3D).
Stakeholders and Interests	• User (Customer): Wants a realistic and detailed 3D preview of glasses frames to assess style and fit before purchase. • Eyewear Retailer: Interested in providing an immersive and accurate preview experience to increase conversion rates. • Developer/Project Owner: Aims to ensure the 3D preview functionality is smooth, detailed, and accessible across devices.
Pre-conditions	• The 3D model of the glasses frames is available and rendered correctly. • The user has selected a specific frame model for preview. • The device supports 3D rendering and interaction (e.g. rotation, zooming).
Success Guarantee	The system successfully renders and displays an interactive high-quality 3D model of the glasses frames for the user to explore.

Main Success Scenario	• User selects a glasses frame from the product catalog. • The system loads the 3D model of the selected frame. • The system renders the frame in a 3D view on the user's device, allowing real-time interaction. • User rotates, zooms, and explores the frame from different angles using touch or mouse controls. • The user can see detailed features of the frame (e.g. material, texture, thickness, design elements).
Extensions	• Poor device performance or 3D rendering issues: The system detects performance issues and switches to a lower-quality rendering mode.
Special Requirements	• Smooth 3D rendering with real-time interaction optimized for different device types (mobile, tablet, desktop). • High-resolution textures and accurate lighting to ensure the frame looks realistic. • Low latency for real-time interactions like zooming and rotating.
Technology and Data Variation	• Uses WebGL, Three.js, or similar frameworks for 3D rendering in browsers. • Trimesh or similar 3D libraries to visualize and manipulate the frame models. • Support for different 3D file formats such as .obj, .glb, and .fbx to ensure compatibility.

Table 2.12: Preview Glasses Frames in 3D Extended Use-Case

Use Case	Fit 3D Frame
Scope	Fitting 3D glasses frames onto the user's face within the virtual try-on platform.
Level	User goal
Primary Actor	User (Customer trying to virtually fit glasses frames).
Stakeholders and Interests	• User (Customer): Wants an accurate representation of how glasses will fit and look on their face in real time. • Eyewear Retailer: Aims to provide a personalized and reliable experience for customers to boost confidence in purchasing glasses online. • Developer/Project Owner: Ensures the system correctly positions and scales the 3D glasses frames onto the user's face.
Pre-conditions	• Face detection and 3D face modeling are fully operational. • The 3D model of the selected glasses frame is available and correctly formatted. • User's device has camera access enabled for the virtual try-on feature.
Success Guarantee	The system accurately fits the 3D glasses model onto the user's face, adjusting for size, positioning, and angle based on real-time facial data.

Main Scenario	• User opens the virtual try-on feature and selects a pair of glasses frames. • The system requests access to the camera and the user grants permission. • The system detects the user's face and constructs a 3D face model using real-time facial tracking and landmark detection. • The system superimposes the selected 3D frame onto the user's face, adjusting for size, angle, and position based on the user's facial dimensions and camera orientation. • The user can move their head or change angles, and the system dynamically adjusts the frame to match the face's movement in real time. • The system provides a visual preview of the frame's fit from various angles, ensuring it aligns correctly with the ears, nose bridge, and eye position. • User can proceed with selecting a different frame or saving the current configuration for further steps (e.g. purchase).
Extensions	• Multiple faces detected: The system notifies the user and asks them to ensure only one face is visible. • Glasses frame does not fit user's face properly: The system automatically adjusts the frame size and position using predefined scaling rules.

Special Requirements	• Real-time face tracking for smooth and dynamic frame fitting. • Accurate scaling and positioning algorithms to ensure frames fit the facial structure properly. • High accuracy for detecting facial landmarks, including the nose bridge, ears, and eye distance to adjust frame placement.
Technology and Data Variation	• Uses MediaPipe Face Mesh or similar facial recognition technology for detecting facial landmarks and creating a 3D face model. • 3D rendering frameworks (e.g. Three.js or WebGL) for dynamically fitting the glasses frames onto the face. • Device differences (e.g. camera quality, processor speed) may impact the accuracy and smoothness of the fitting process.

Table 2.13: Fit 3D Frame Extended Use-Case

Use Case	Customize Frame
Scope	Customization of 3D glasses frames within the virtual try-on platform.
Level	User goal
Primary Actor	User (Customer personalizing their selected glasses frames).

Stakeholders and Interests	• User (Customer): Wants to personalize their glasses frame by adjusting color, material, and design details to reflect their style preferences. • Eyewear Retailer: Interested in providing customization options to increase customer satisfaction and purchase likelihood. • Developer/Project Owner: Aims to offer a seamless and intuitive customization experience that allows users to modify frame elements in real time.
Pre-conditions	• The 3D model of the glasses frame is available in the system. • The customization interface is loaded and operational. • Users have selected a frame they want to customize.
Success Guarantee	The system allows the user to easily and accurately modify the appearance of the 3D glasses frame and the changes are reflected in real time.
Main Success Scenario	• User selects a glasses frame from the catalog. • The system loads the 3D model of the selected frame. • The system presents a customization interface with various options (e.g., color, size). • User chooses a color from the palette, and the system updates the frame in real time to reflect the new color. • The system updates the 3D model in real time to show all customizations. • User can reset changes, save the customized frame, or proceed to virtual try-on.

Extensions	• Performance issues with real-time rendering: The system reduces the resolution or level of detail in the customization view to maintain smooth performance.
Special Requirements	• Real-time updates and high-quality rendering of changes to ensure users can visualize their customizations immediately. • Flexibility to support different customization options like colors and size alterations. • User-friendly interface to allow for intuitive changes without requiring advanced design skills.
Technology and Data Variation	• Uses WebGL, Three.js, or similar 3D rendering technologies for real-time updates to the frame model. • The customization data (color, size) is stored and applied to the 3D mesh dynamically.

Table 2.14: Customize Frame Extended Use-Case

Use Case	Integrate with Retailers
Scope	Integration of the virtual glasses try-on system with retailers' platforms (e.g., e-commerce websites or physical store systems).
Level	System-Level
Primary Actor	Retailer (Eyewear retailers looking to integrate the virtual try-on solution into their platform).

Stakeholders and Interests	• Retailer: Wants to offer customers a seamless virtual try-on experience within their own platform to increase online sales and customer engagement. • Customer: Interested in trying on glasses virtually before making a purchase through the retailer's platform. • Developer/Project Owner: Ensures the integration is smooth and scalable, supporting multiple retailers without requiring significant changes to the system. • Eyewear Manufacturers: Want their glasses to be accurately represented in the virtual try-on environment.
Pre-conditions	• The retailer has a functioning e-commerce platform or digital system that can accommodate the integration. • The retailer's platform is able to handle API requests and manage the necessary product data (e.g., 3D glasses models, product catalog information). • The virtual try-on solution provides an API for seamless integration with external platforms.
Success Guarantee	The virtual try-on system is successfully integrated into the retailer's platform, providing their customers with a smooth and accurate try-on experience leading to higher engagement and conversion rates.

Main Success Scenario	• The retailer contacts the developer or accesses the API for the virtual try-on system. • The developer provides documentation and technical support for integration, including API endpoints, authentication, and data formats. • The retailer's development team integrates the try-on solution by embedding it into the existing platform (e.g., through API calls). • The retailer uploads or links their glasses catalog and 3D models to the try-on system. • Customers visiting the retailer's platform can access the virtual try-on feature when viewing a specific pair of glasses. • The system fetches the necessary data (e.g., 3D model of the frame, user face detection) and enables the customer to virtually try on glasses. • The try-on session data (e.g., selected frame customization options) is integrated with the retailer's purchase flow, allowing customers to add the glasses to their cart and complete the transaction.
Extensions	• API connection failure or incorrect integration: The system notifies the retailer's development team of the issue and provides troubleshooting guidance.

Special Requirements	• The integration must be flexible enough to support various types of retailer platforms (e.g., websites, mobile apps). • High availability and scalability of the API to handle large volumes of customer requests, especially during peak periods.
Technology and Data Variation	• The system provides RESTful API endpoints for integration with various platforms. • Different data formats (e.g., JSON for catalog data, 3D model formats like .obj or .glb) may be required to align with the retailer's systems.

Table 2.15: Integrate with Retailers Extended Use-Case

Use Case	Select Frame Glasses
Scope	Allow users to browse and select glasses frames for virtual try-on or purchase.
Level	User goal
Primary Actor	User (Customer selecting glasses frames).
Stakeholders and Interests	• User (Customer): Wants to easily browse and select glasses frames that suit their style, size, and preferences for a virtual try-on or purchase. • Retailer: Wants to ensure users can seamlessly browse and select frames, encouraging higher engagement and conversions. • Developer/Project Owner: Aims to provide an intuitive and responsive interface for frame selection.

Pre-conditions	• The system has a catalog of available glasses frames, including product details and 3D models. • The user has access to the platform, either on the web or through a mobile app.
Success Guarantee	The user is able to browse and select a glasses frame from the available options and the selection is accurately registered in the system.
Main Success Scenario	• User logs into the platform and navigates to the glasses selection section. • The system presents a catalog of glasses frames. • User browses the catalog and selects different frames to view in more detail. • The system displays additional information about the selected frame, including images, descriptions, and a 3D model. • User selects a frame for virtual try-on. • The system confirms the frame selection and prepares the model for virtual try-on. • The user can continue browsing or try the frame virtually.
Extensions	• Frame customization available: The system allows the user to customize certain aspects of the frame (e.g., color) before selecting it for virtual try-on or purchase.

Special Requirements	• Quick loading of frames and information to enhance user experience. • The interface must be intuitive, allowing for easy navigation between different frames.
Technology and Data Variation	• Integration with a product database to manage and display the frames.

Table 2.16: Select Frame Glasses Extended Use-Case

2.2 Functional Requirements

2.2.1 Face 3D Model Generation

Following are the requirements for module 1:

1. **Face Detection and Identification:** The system shall detect and identify faces in video input using facial features (eyes, nose, mouth, jawline). [4]
2. **Feature Extraction:** The system shall use facial landmarks to categorize and extract facial features.
3. **3D Mesh Generation:** The system shall generate a 3D mesh of the face using the extracted features.
4. **Depth Estimation:** The system shall estimate the size of the user's face using depth estimation via the diameter of the iris. [4]

2.2.2 Glasses 3D Model Generation

Following are the requirements for module 2:

1. **Image Preprocessing:** The system shall preprocess images to segment and separate the glasses from the background. [1]
2. **3D Mesh Creation:** The system shall create a 3D mesh of the glasses using multiple images from different angles through photogrammetry techniques. [2]

2.2.3 3D Model Integration

Following are the requirements for module 3:

1. **Feature Extraction:** The system shall identify and extract common features from the glasses and the face to ensure compatibility.
2. **Feature Mapping:** The system shall map the features of the glasses to the features of the face for integrating.
3. **Mesh Integration:** The system shall connect and integrate the glasses mesh with the face to create a integrated model.

2.2.4 Glasses Customization

Following are the requirements for module 4:

1. **Color Selection:** The system shall allow users to select from a variety of colors for the glasses frame.
2. **Size Adjustment:** The system shall enable users to adjust the size of the glasses to fit the individual.

2.2.5 API Development

Following are the requirements for module 5:

1. **API Endpoints:** The system shall provide API endpoints for each functionality of the virtual glasses try-on system.
2. **Authentication Measures:** The system shall implement authentication on all API endpoints, to make sure that only authorized users can access them. [5]

2.2.6 Client Web Application Development

Following are the requirements for module 6:

1. **Catalog Management:** The web application shall allow users to view and manage a catalog of available glasses.
2. **Interactive 3D Previewer:** The web application shall include an interactive 3D previewer to allow users to visualize the glasses from all angles.
3. **Virtual Try-On:** The web application shall enable users to virtually try on all available glasses using the integrated API, displaying the customized 3D model.

2.3 Non-Functional Requirements

2.3.1 Performance

1. **Low Latency:** The system shall process video input and give video output, with a maximum latency of 500 milliseconds.
2. **Smooth Frame Rate:** The system shall support a frame rate of at least 30 frames per second for a smooth video viewing experience.

2.3.2 Reliability

1. **Uptime:** The system shall have an uptime of 99% to ensure it is consistently available for users.
2. **Error Handling:** The system shall implement error handling to easily recover from failure without loss of data.

2.3.3 Usability

1. **Intuitive Interface:** The system shall provide an intuitive user interface, so no more than three clicks will be required to access the virtual try-on feature.
2. **Tooltips:** The application shall include tooltips to help users in navigating through the interface.

2.3.4 Security

1. **Secure Coding Practices:** Secure coding practices will be used to prevent vulnerabilities (e.g., SQL injection and cross-site scripting).
2. **Data Encryption:** The user data stored in the system shall be encrypted to protect it from unauthorized access.

Chapter 3

System Overview

Functionality:

OptiFit is a software solution that enables users to virtually try on eye wear (glass frames) using their device's camera. It uses advanced facial recognition and 3D modeling techniques to superimpose 3D models of glass frames onto the user's face in real-time. The solution includes several key features like Frame selection, Virtual Try-on, Customization, Glasses 3D model creation, face 3D model creation, integration with retailers, and 3D previewer.

Context:

OptiFit addresses the challenges faced by eyewear retailers and customers when it comes to purchasing glasses online. In a traditional retail environment, customers try on physical frames to ensure they match their face shape and style. However, the rise of online shopping means customers are increasingly making eyewear purchases without being able to try them on.

To bridge this gap, OptiFit provides a virtual try-on experience that allows customers to make informed decisions from the comfort of their homes. This increases customer satisfaction and reduces the chances of returns or dissatisfaction after purchase.

Design:

The project is divided into several components and layers:

1. Application Layer (Frontend)

- Handles user interactions, including the selection of glasses, camera access, and virtual try-on features.
- Provides an intuitive user interface that allows users to view glasses from different angles, adjust sizes, and customize frames.

2. Business Layer (Backend)

- Manages core business logic, including facial recognition, 3D model generation, and API integration.
- Processes data from the user's face and the glasses 3D models and merges them in real time.
- Handles frame customization requests and stores frame catalogs, user data, and transaction details.

3. Data Layer

- Consists of databases that store user information, glasses frame details, 3D models, and transaction records.
- Supports non-relational data storage to manage various formats (e.g., user preferences, frame images, and 3D model files).

3.1 Architectural Design

3.1.1 Architecture Diagram

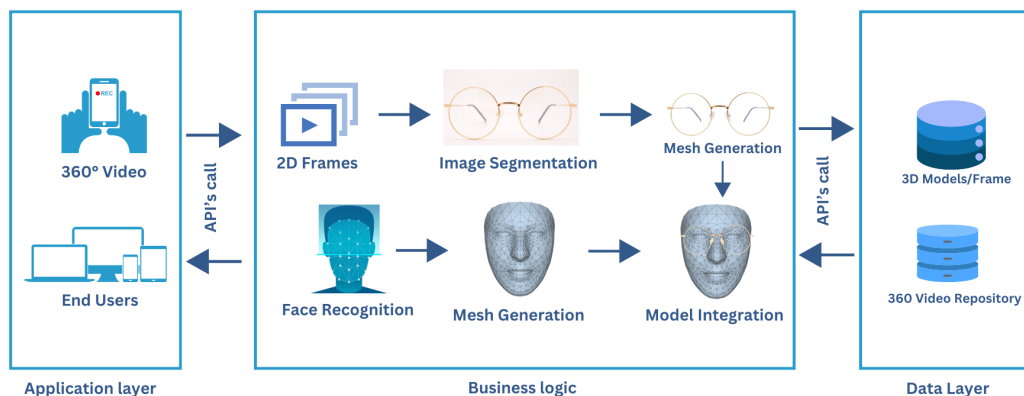


Figure 3.1: Architecture diagram

3.2 Design Models

3.2.1 Activity Diagram

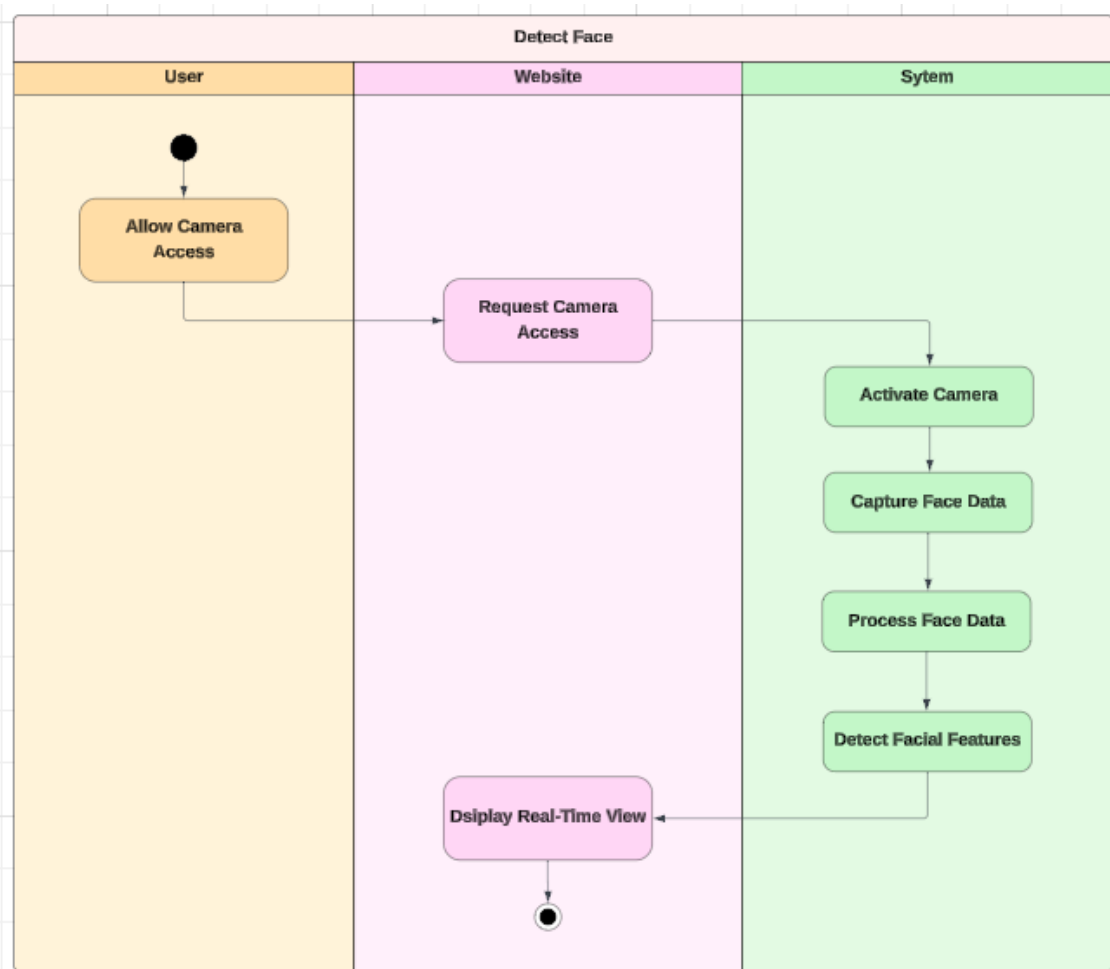


Figure 3.2: Detect Face Activity Diagram

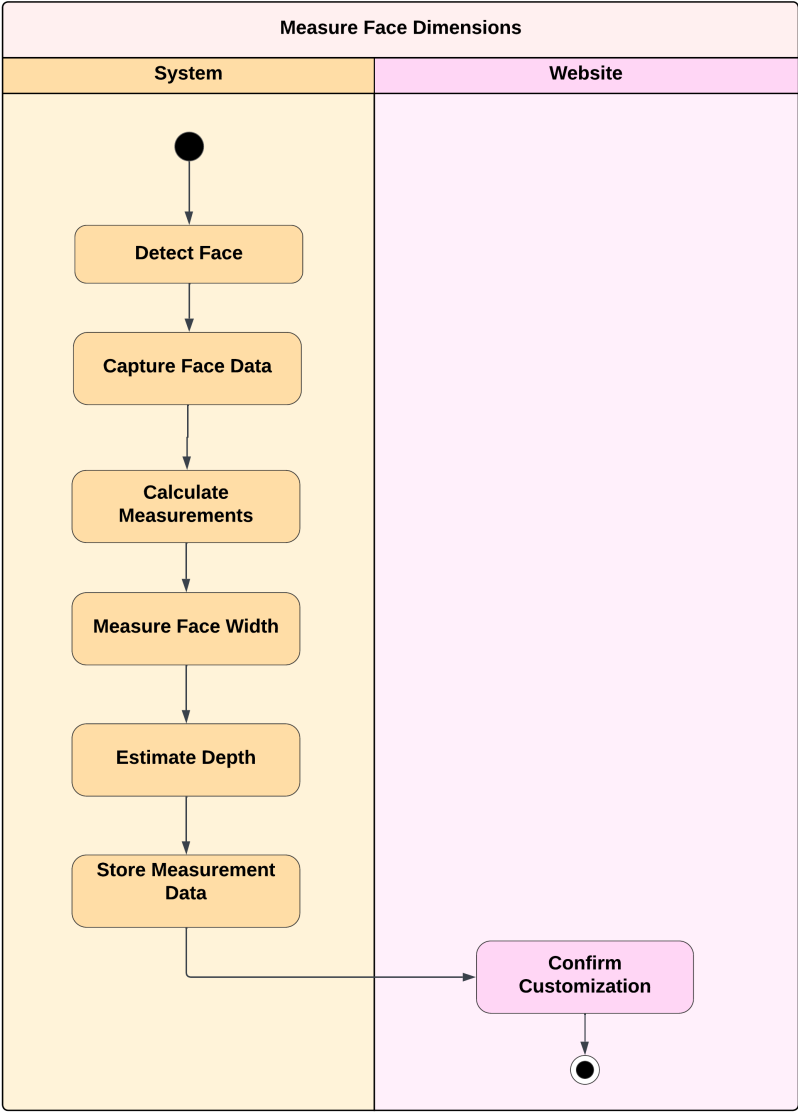


Figure 3.3: Measure Face Dimension Activity Diagram

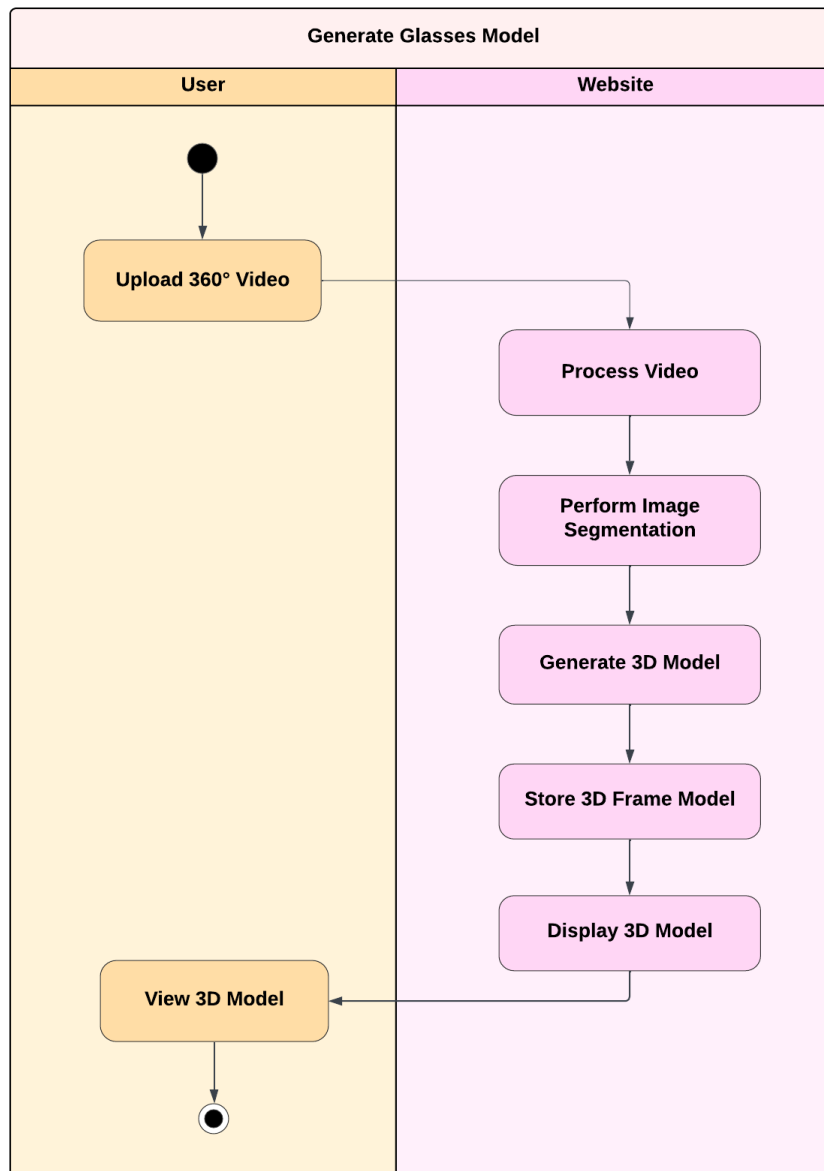


Figure 3.4: Generate Glasses Model Activity Diagram

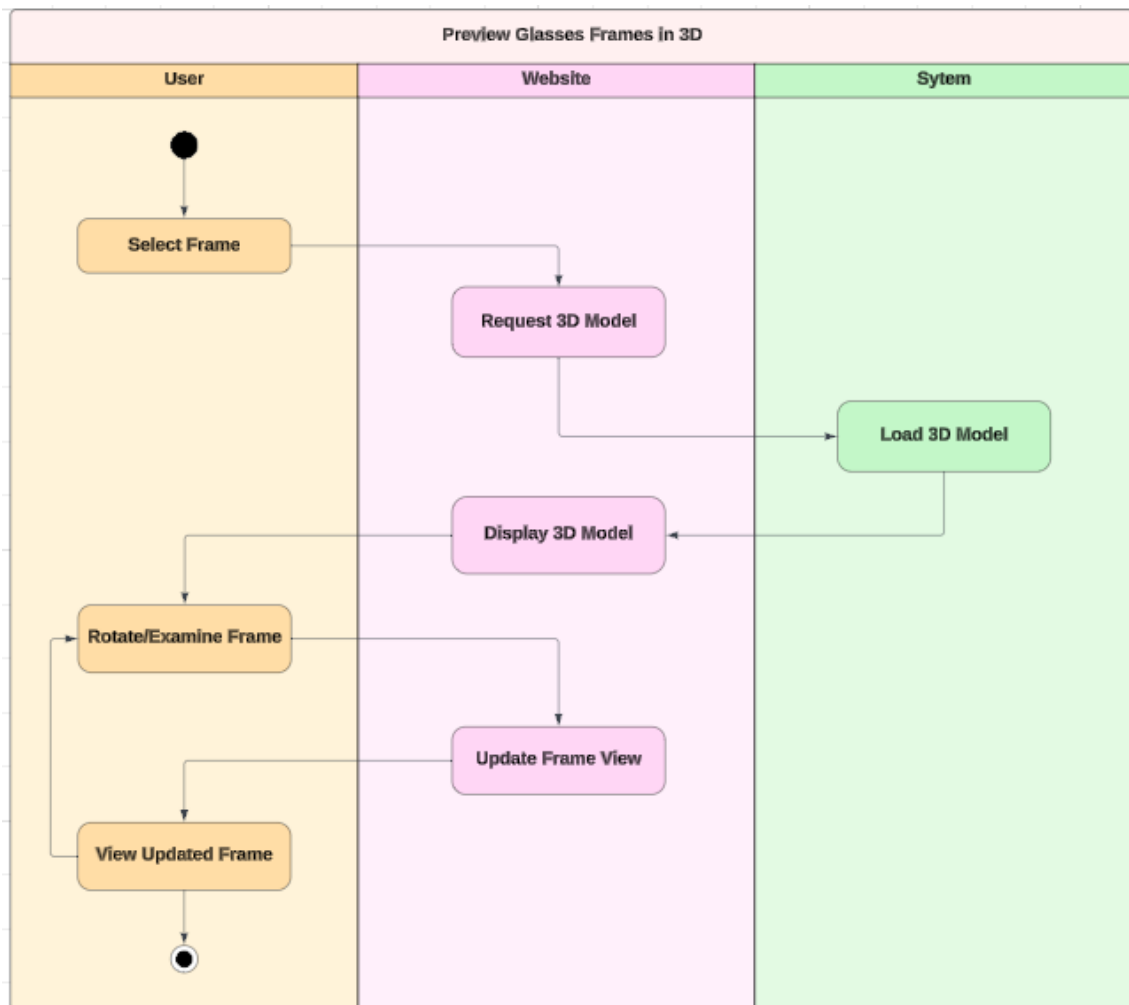


Figure 3.5: Preview Glasses Frames in 3D Activity Diagram

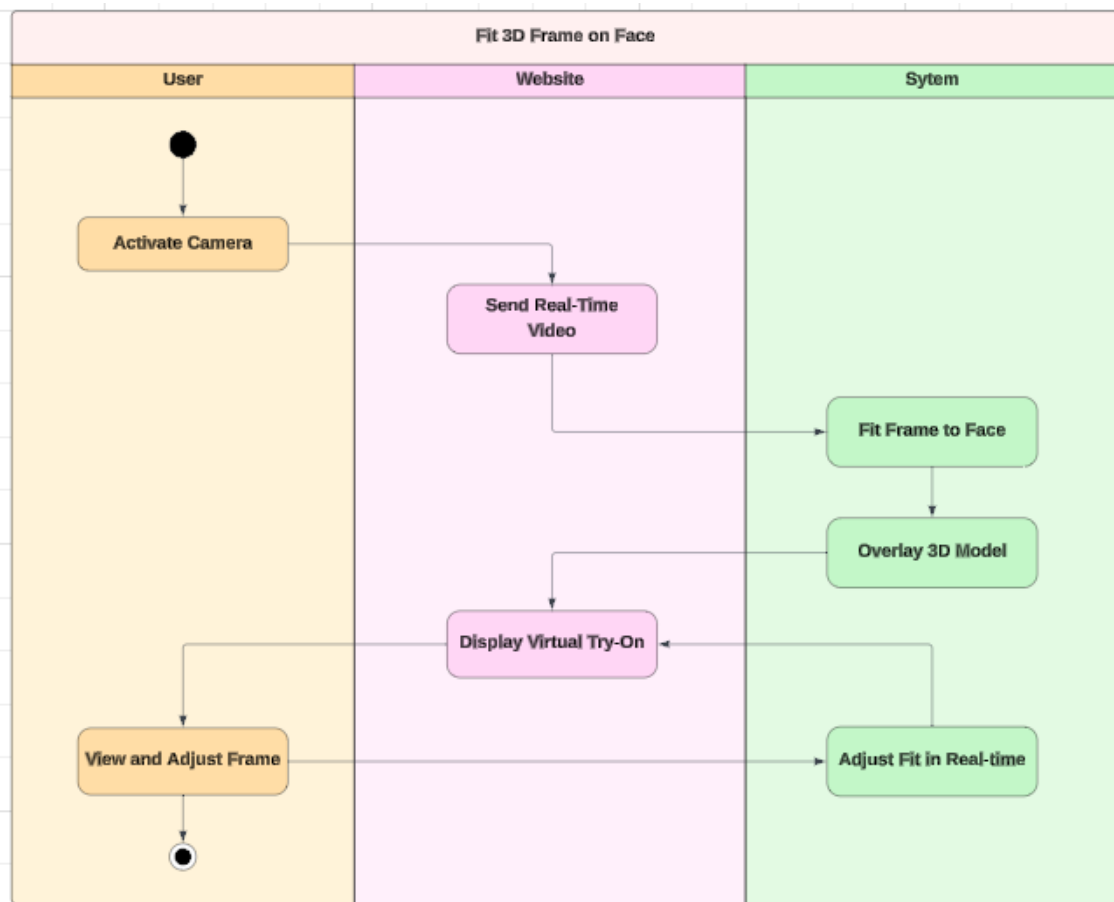


Figure 3.6: Fit 3D Frame on Face Activity Diagram

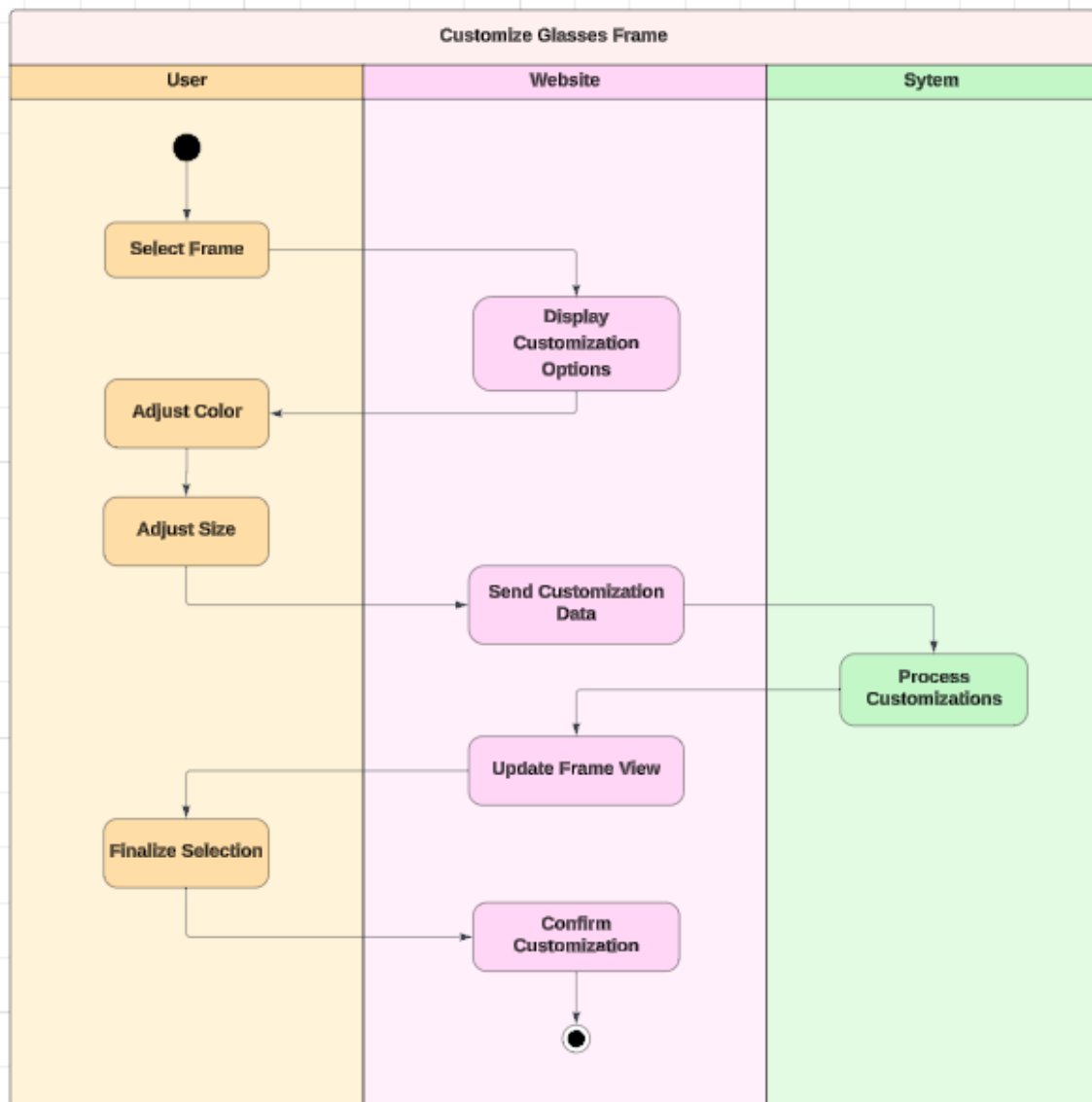


Figure 3.7: Customize Glasses Frame Activity Diagram

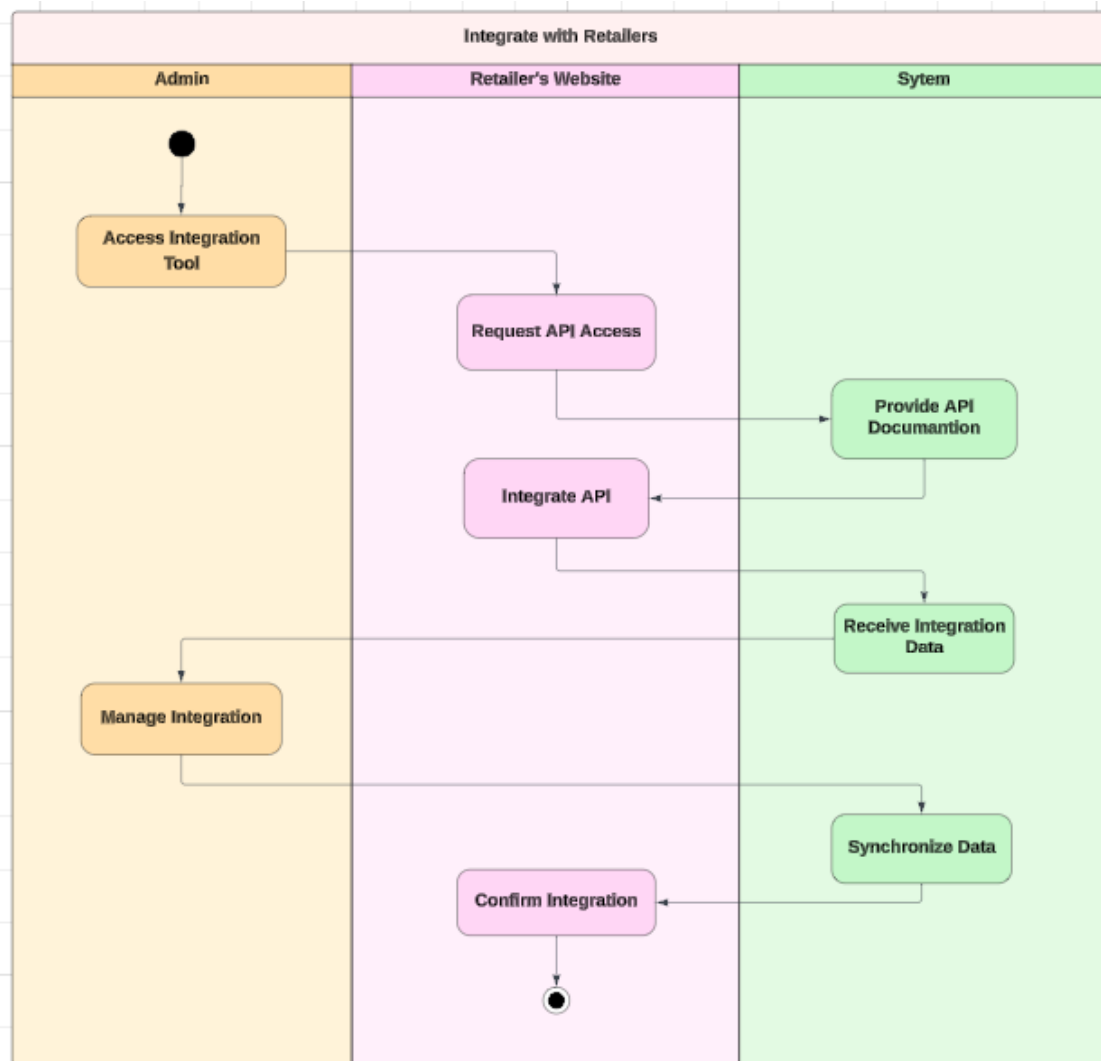


Figure 3.8: Integrate With Retailer Activity Diagram

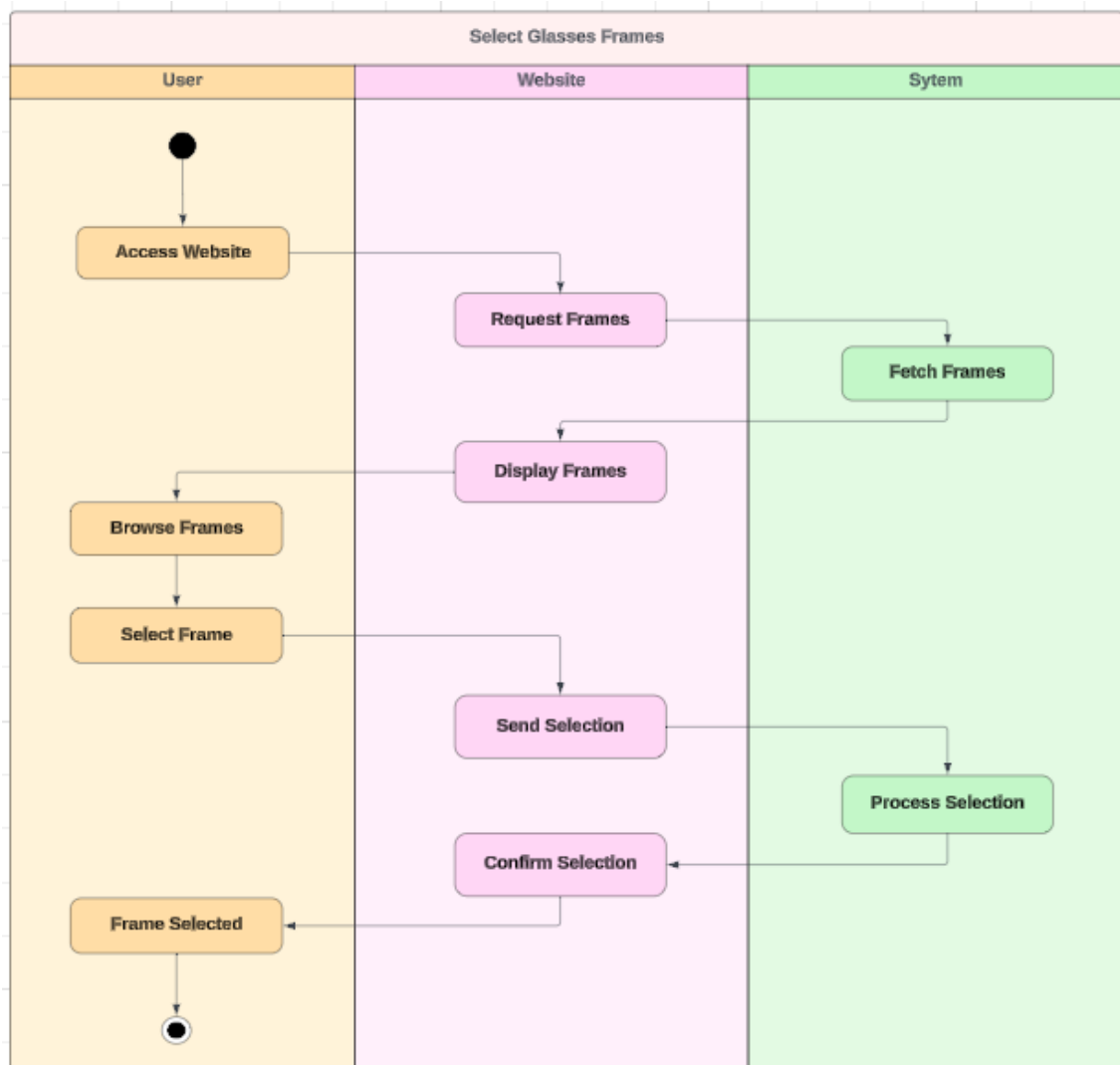


Figure 3.9: Select Glasses Frame Activity Diagram

3.2.2 Data Flow Diagram

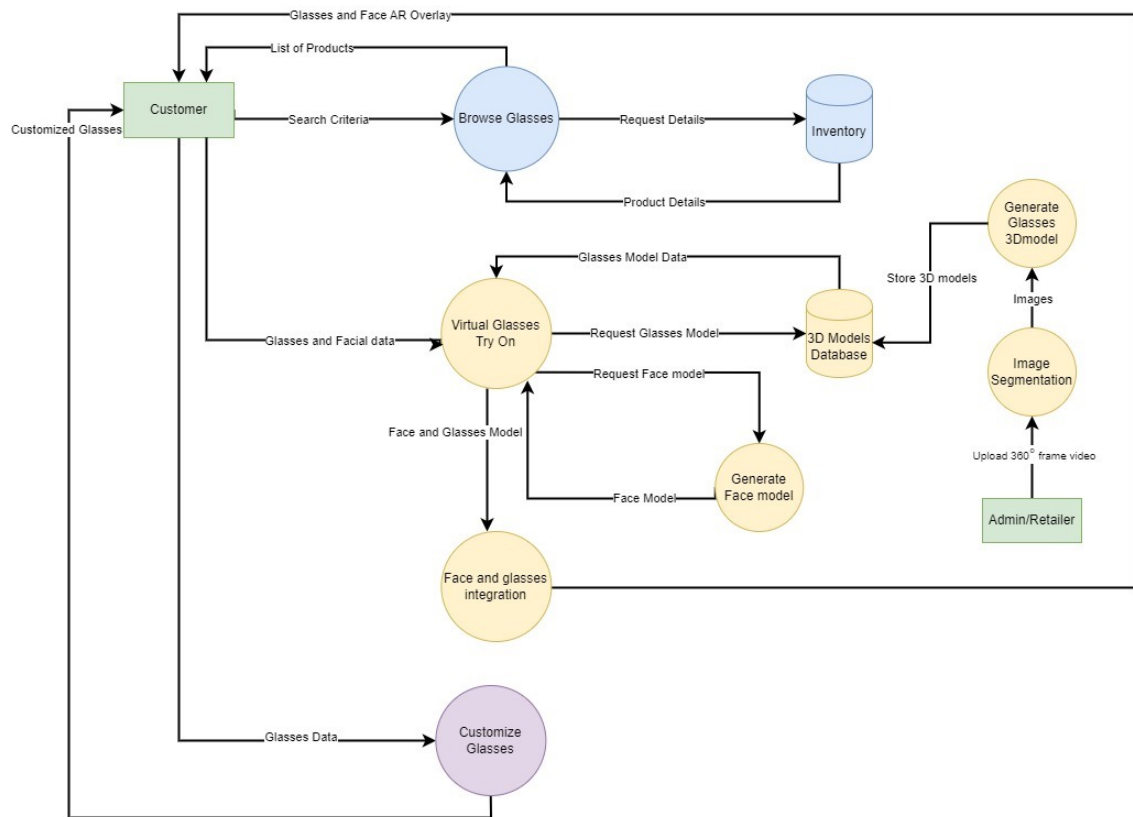


Figure 3.10: Data Flow Diagram

3.2.3 System-level Sequence Diagram

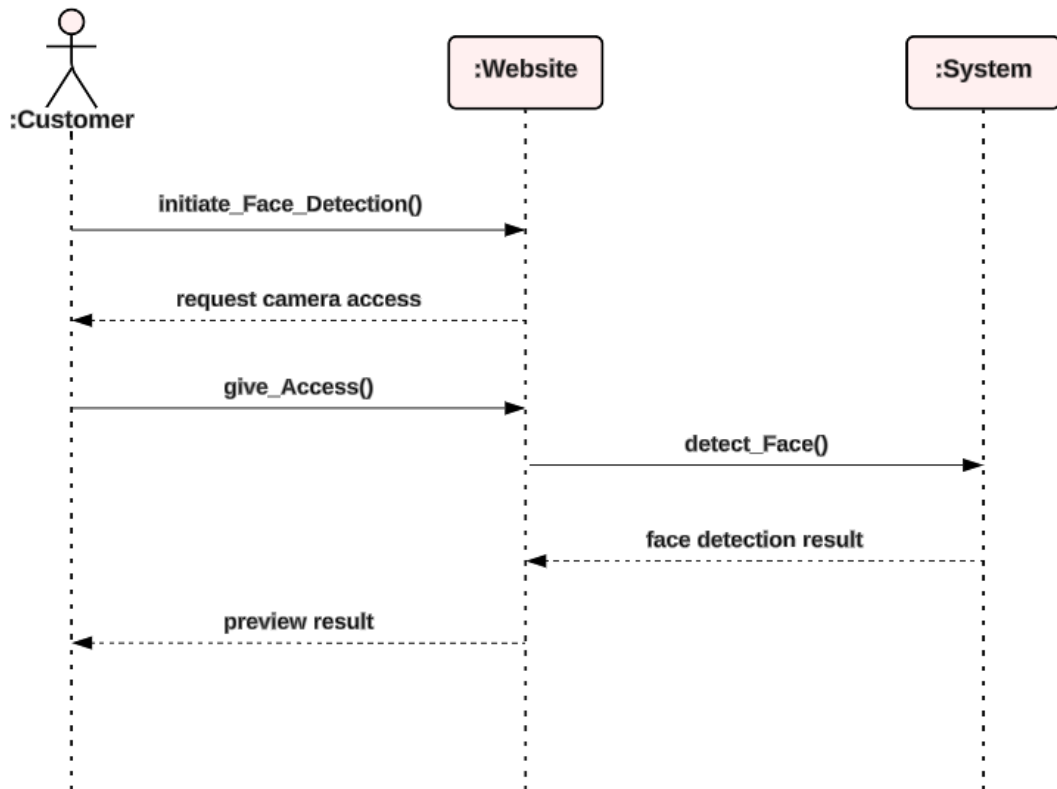


Figure 3.11: Detect Face

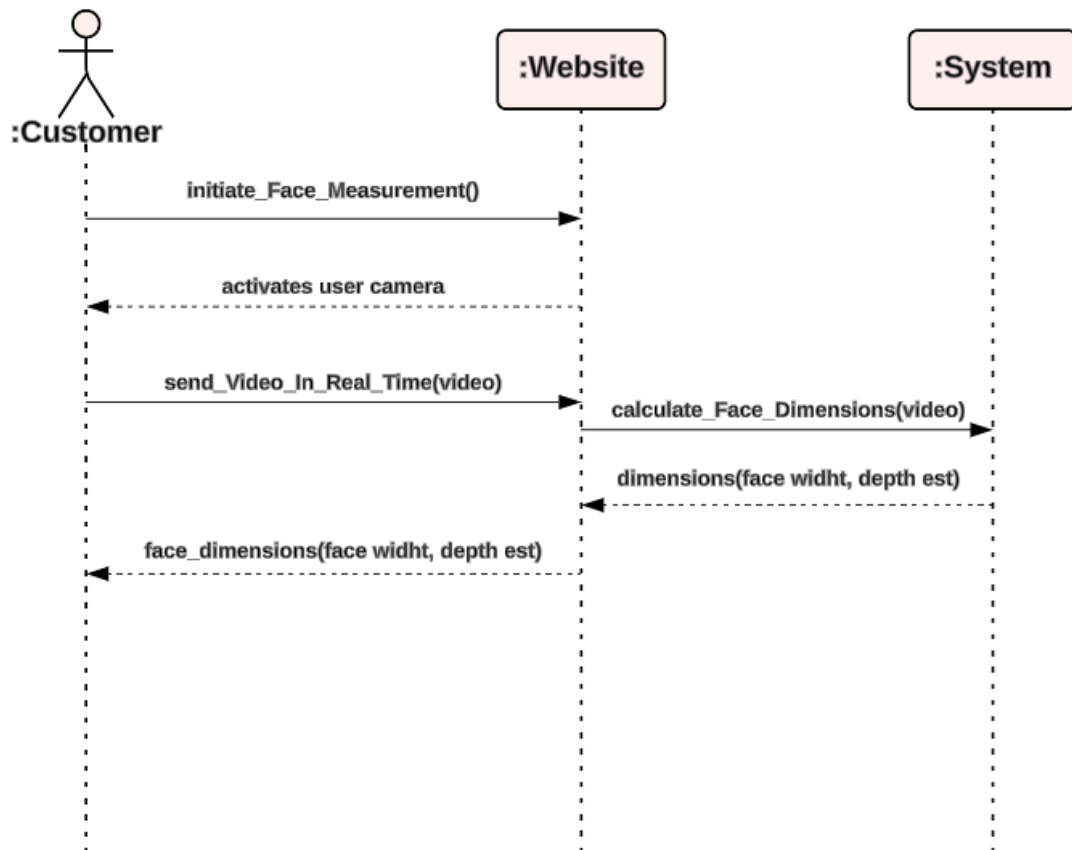


Figure 3.12: Measure Face Dimensions

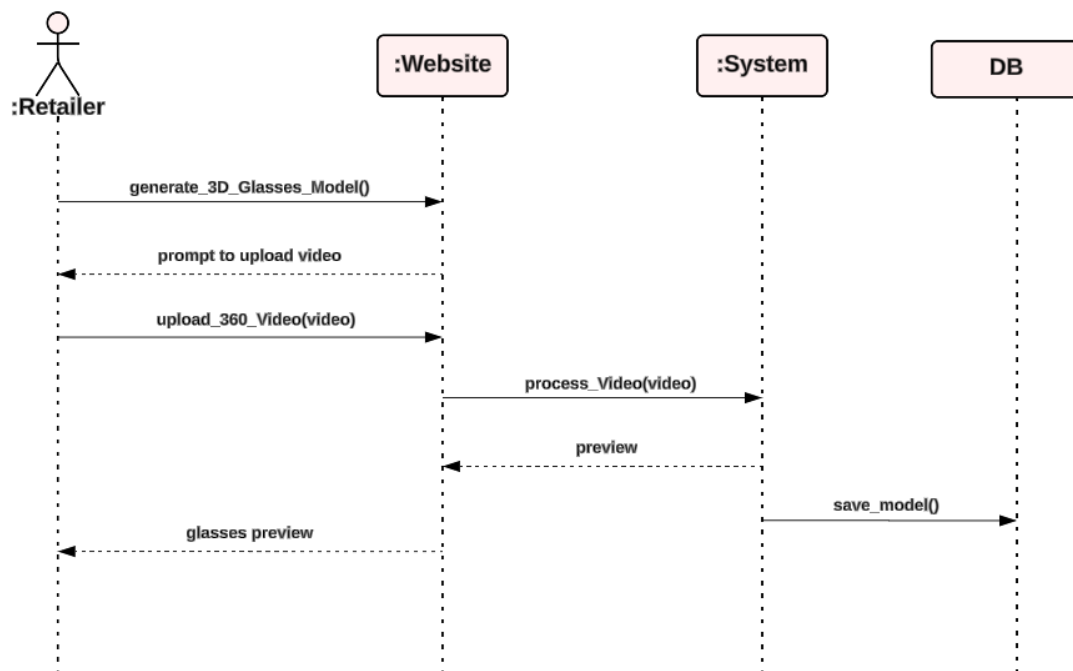


Figure 3.13: Generate Glasses Model

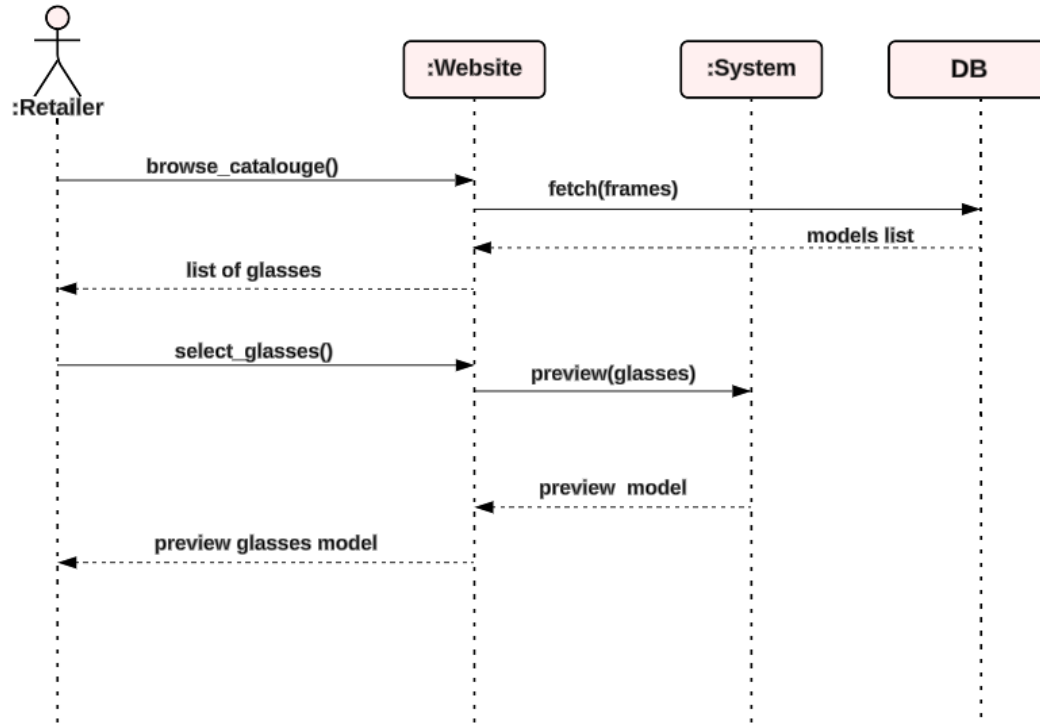


Figure 3.14: Preview Frames in 3D

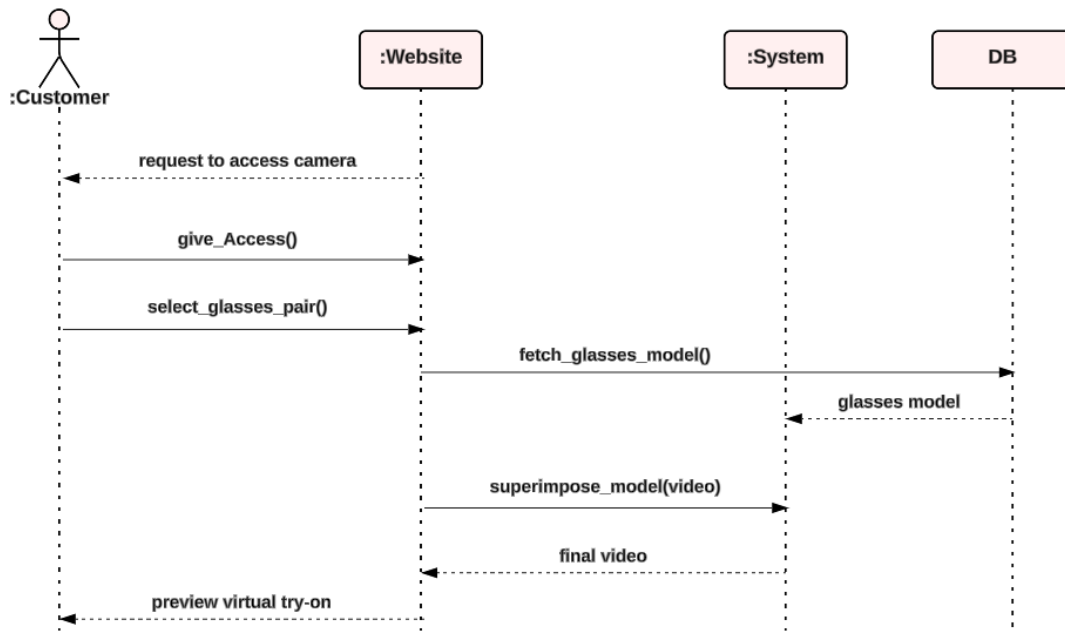


Figure 3.15: Fit 3D Frame

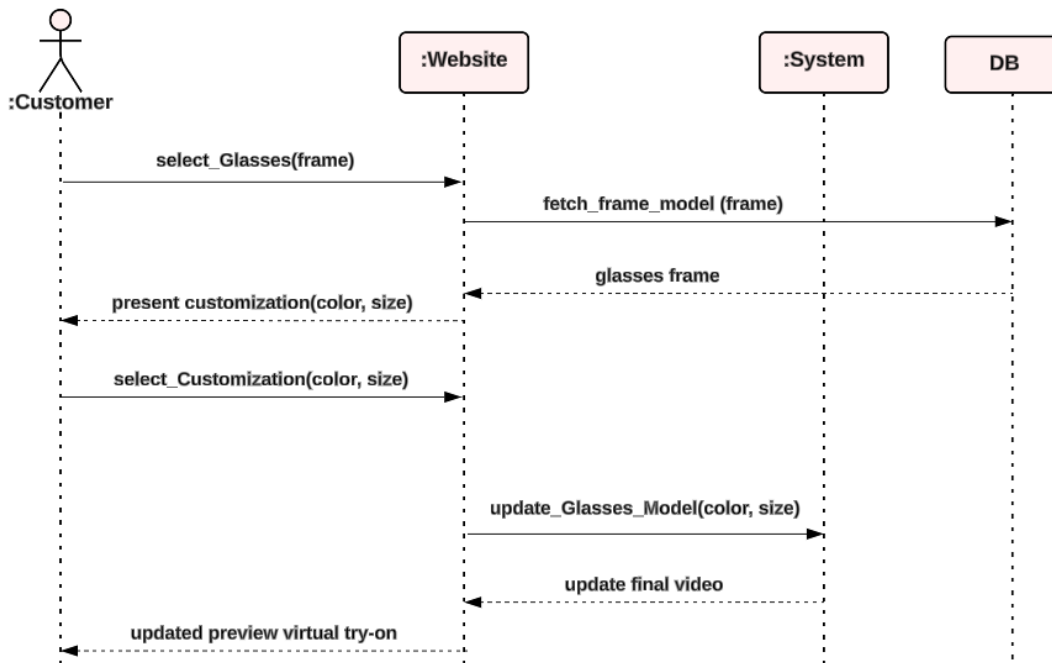


Figure 3.16: Customize Glasses Frame

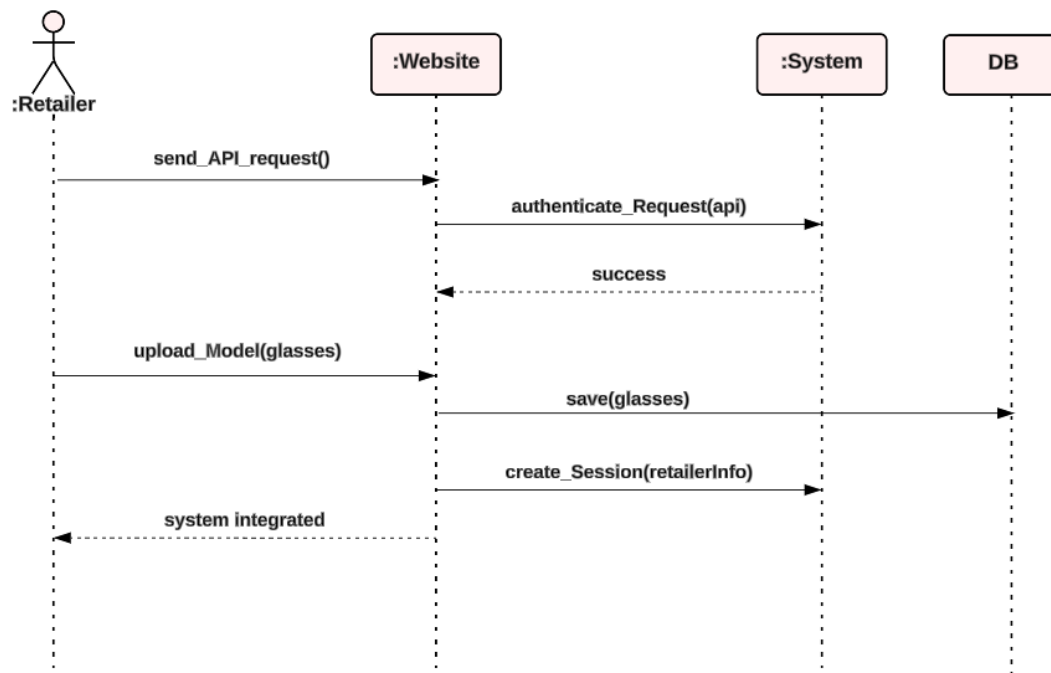


Figure 3.17: Integrate with Retailers

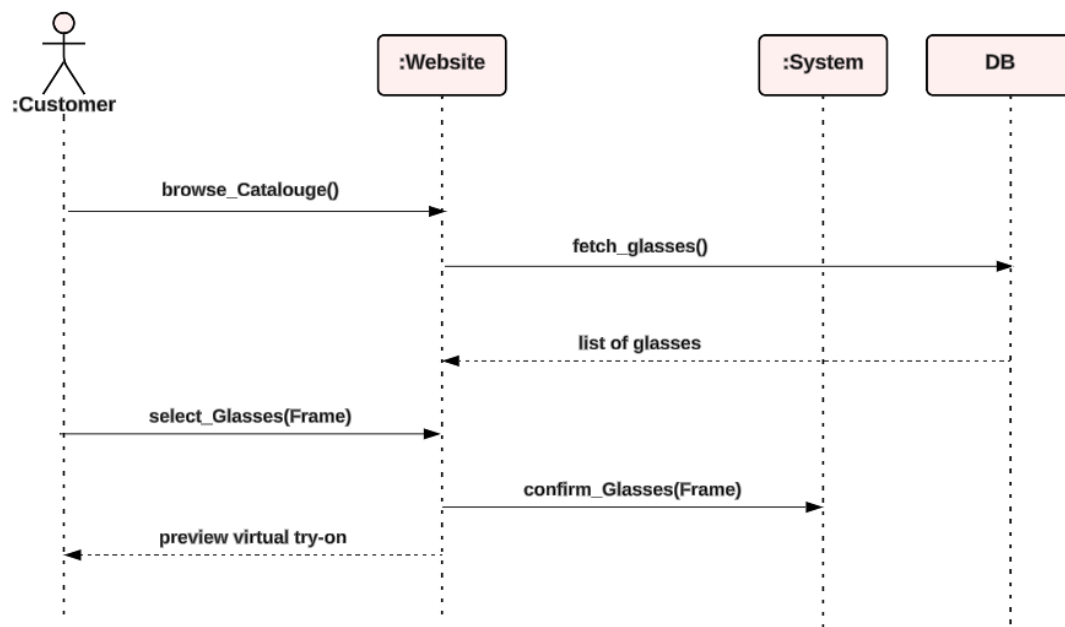


Figure 3.18: Select Glasses Frames

3.2.4 ERD Diagram

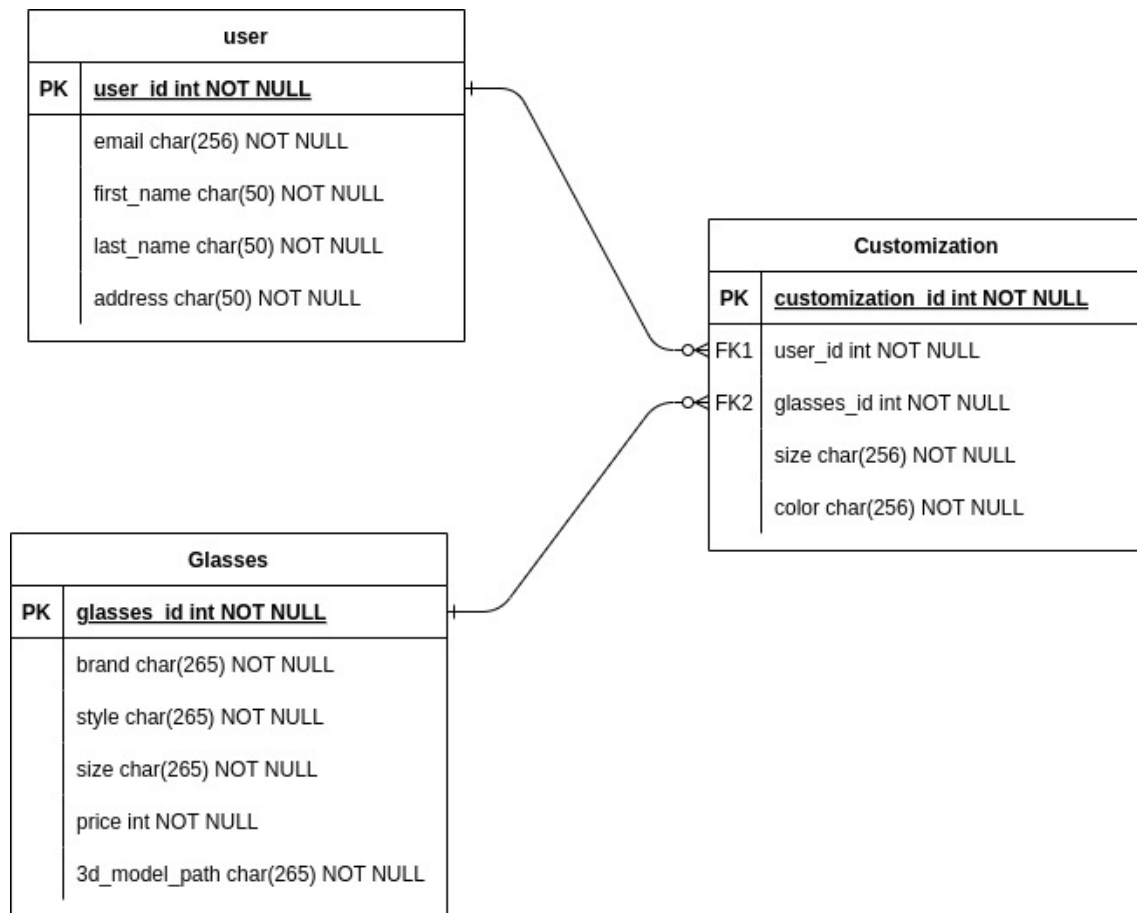


Figure 3.19: ERD Diagram - OptiFit

3.3 Data Design

The information domain of our project is transformed into structured data such as image frames, 3D meshes, point clouds, and JSON objects. The data structures will be stored in databases and processed through processes like 3D model generation, customization, and API interaction. Our system uses file-based storage for 3D models and document-based databases to store user preferences and glasses data.

Bibliography

- [1] Coralie Bernay-Angeletti and Jean-Marie Favreau. Technical report : 3d mesh segmentation application to glasses. *Hal.science*, Oct 2017.
- [2] Sorawich Pongpiyapaiboon. An introduction to photogrammetry: How to create 3d models from photos. *SoraScience Blog*, December 30 2023.
- [3] Straits Research. Eyewear market size, share trends analysis report by product. 2022.
- [4] Andrey Vakunov and Dmitry Lagun. Mediapipe iris: Real-time iris tracking & depth estimation. *Google Research Blog*, August 6 2020.
- [5] Mirko Zorz. Overlooked essentials: Api security best practices. *Help Net Security*, July 17 2024.